

Mitigating Compute Dilution in Compute-Constrained Online RGB-D 3D Gaussian Splatting

via Dual Map-Growth Throttling and Sensor-Adaptive Routing

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Abstract

Dense online reconstruction from streaming RGB-D sensors under a rigid per-frame compute budget (15 ms optimize budget) is bottlenecked by a structural failure mode we call **compute dilution**: uncontrolled densification floods the Gaussian map with immature primitives ($\sim 100k$ splats in 150 frames), so fixed-budget gradient steps are spread over ever more under-converged splats while mature surface geometry starves. Conventional wisdom blames candidate *ranking* — which Gaussians to optimize — and proposes learned utility predictors. Through exact counterfactual measurement we show ranking is secondary: 21.37% of isolated Gaussian updates have *negative* marginal utility ($U_i^* < 0$, rising to 23.86% in the top-error decile), joint updates are severely non-additive ($R_{\text{add}}(16) \approx 0.005$), and an offline-learned utility model with rank signal ($\rho \approx 0.20$) degrades closed-loop fidelity by -2.51 dB. The binding constraint is the map-growth process itself. We introduce **dual map-growth throttling** — coverage-saturation throttling ($\kappa \geq 0.90$) plus unrefined-backlog backpressure — which holds the map at 23k vs 88k primitives and directs the fixed budget at mature refinement. On TUM `fr2.xyz` (150 frames $\times$ 8 seeds) this reaches 27.67 dB mean PSNR: **+2.63 dB** over the strongest selective baseline and **+4.62 dB** over an RTG-SLAM-policy baseline ($p_{\text{Holm}} = 0.0391$), recovering **98.0%** of headroom to the unconstrained ceiling (0.18 dB gap). We further show every *fixed* rule overfits its sensor regime: the same pipeline loses -3.32 dB ($p = 0.0078$) on clean synthetic Replica, where raw-error selection wins instead. A **sensor-adaptive router** driven by raw depth hole-rate (TUM ~ 0.25 vs Replica ~ 0.000 , 300–500 $\times$ separation, 5/5 scenes) selects the winning rule per substrate and wins on both. All benchmarks use matched budgets, paired Holm-corrected tests, and frozen reproduction artifacts.

1 Introduction

3D Gaussian Splatting (3DGS) [1] has become the dominant primitive for photorealistic dense mapping, driving online RGB-D systems such as SplatAM [2], Point-SLAM [3], MonoGS [4], Photo-SLAM [5] and RTG-SLAM [6]. De-

ployed on robots and headsets, these systems face a hard constraint the offline literature ignores: each streaming frame admits only milliseconds of optimization. The standard response is *selection* — rank Gaussians by residual error, influence, or a learned utility score, and optimize the top subset. Implicit in this response is an assumption we challenge: that reconstruction quality under budget is determined by *which* Gaussians are picked.

We argue the dominant effect is *how many* Gaussians compete for the budget. Online pipelines spawn splats whenever rendering error exceeds a threshold; on real sequences the map grows past 100k primitives within 150 frames. Because each new splat is born unrefined, it both demands optimization and degrades renders until converged — the fixed budget is diluted across an ever-larger pool of ever-younger primitives while converged surfaces stop receiving updates. We call this **compute dilution**, and we show it explains more of the quality gap than any ranking rule: removing growth control while keeping the best ranker loses far more than removing the ranker while keeping growth control (Sec. 6).

Three empirical findings support this reframing. First, residual error does not imply positive optimization return: counterfactual single-Gaussian interventions degrade global quality in 21.37% of cases. Second, Gaussian utilities do not compose: groups of 16 jointly-optimized splats realize 0.5% of the sum of their isolated gains, so pointwise ranking cannot be a combinatorial optimum and contextual re-ranking buys nothing measurable (rank stability $\bar{\rho} = 0.8916$ at 424 $\times$ overhead). Third, offline-learned utility prediction does not transfer: moderate offline rank correlation reverses to -2.51 dB closed-loop under trajectory shift and gradient coupling. Together these rule out the “better ranker” research direction and point at the growth process.

Our response has two parts. **Dual map-growth throttling** regulates spawning with (i) a coverage-saturation knee ($\kappa_t \geq 0.90$ measured by transmission) that ramps new-primitive allowance $1.0 \rightarrow 0.20$, and (ii) backlog backpressure that penalizes densification while more than 500 primitives await their first refinements. **Sensor-adaptive routing** addresses the second discovery of this work: fixed rules overfit their sensor-noise regime. The identical throt-

tled pipeline that gains +3.59 dB on noisy Kinect data loses −3.32 dB on clean synthetic data, where plain raw-error selection dominates. A training-free router keyed on raw depth hole-rate — missing measurements, i.e. sensor physics rather than scene content (TUM 0.18–0.31 vs Replica $\approx$ 0.000) — picks the winning rule per substrate and wins on both, while every fixed policy fails one substrate badly.

Contributions. (1) A budgeted subset-selection formulation with an exact counterfactual oracle, exposing negative utility and non-additivity. (2) Dual coverage/backlog throttling reaching 98.0% of unconstrained headroom with a $3.8\times$ smaller map (8 seeds, Holm-significant). (3) Sensor-adaptive routing from raw hole-rate with 5/5 scene separation and significant wins on both substrates. (4) Fully reported negative results — learned utility, contextual re-ranking, synthetic failure, and two failed noise features — with frozen artifacts for every claim.

2 Related Work

Online 3DGS SLAM and dense mapping. SplatAM [2] established silhouette-guided differentiable SLAM with 3D Gaussians; Point-SLAM [3] uses dynamic neural point clouds; MonoGS [4] and Photo-SLAM [5] push toward real-time operation with loop closure; RTG-SLAM [6] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 7). **Budgeted and utility-driven selection.** Error-only, error $\times$ influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers, and we show their ordering flips across sensor regimes — the motivation for adaptivity. **Submodular maximization and knapsack allocation.** $\max_S \Delta Q(S)$ s.t. $C(S) \leq B$ is a knapsack instance related to submodular maximization with greedy $(1 - 1/e)$ guarantees [7]. Our measured $R_{\text{add}} \ll 1$ shows the objective is far from additive in practice, which is precisely why we regulate the ground set (map growth) instead of refining the greedy order.

3 Problem Formulation

Let $G_t = \{g_1, \dots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t . Differentiable alpha compositing renders $\hat{I}(u) = \sum_i c_i \alpha_i \mathcal{G}_i(u) \prod_{j < i} (1 - \alpha_j \mathcal{G}_j(u))$ (and depth analogously). Per-frame scheduling solves

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \quad (1)$$

with decoupled cost $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ for K microsteps and ΔQ measured by exact full-scene counterfactual renders (active edits embedded in all N splats, avoiding the 31% gradient error of affine-background approximations). The oracle $U_i^* = \Delta Q_i^{\text{exact}} / \Delta C_i$ is our measurement instrument, never a runtime input. It reveals: (i) negative utility in 21.37% of interventions (23.86% top-error decile) — high error marks silhouettes, discontinuities, and coupled layers where gradient steps hurt; (ii) non-additivity $R_{\text{add}}(4) \approx -0.358$, $R_{\text{add}}(16) \approx 0.0053$, so pointwise top- K is a reference, not a bound, and greedy conditional oracles ($\arg \max_{i \notin S} \Delta Q(i | S) / \Delta C(i | S)$) coincide with static greedy in selection outcome (conditional advantage 0); (iii) offline TwoHeadMLP rank signal ($\rho \approx 0.20$, NDCG@20 0.46) reverses to −2.51 dB closed-loop.

4 Method

Fast Approximate Attribution (FAA). Exact Gaussian-to-pixel attribution costs 523.9 ms/frame. Center-sampled projection ($\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4} e(\lfloor \mu_{2D,i} \rfloor)$) cuts this to **0.85 ms** ($614.5\times$) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. **Selection.** Candidates are scored by error $\times$ influence mass with temporal EMA filtering ($\beta = 0.90$) and packed by fractional-knapsack value density under $B_t = 15$ ms with safety factor 1.10. **Dual throttling.** Coverage κ_t = fraction of valid pixels with transmission < 0.5 . The per-frame spawn cap ramps $1.0 \rightarrow 0.20$ over $\kappa_t \in [0.80, 0.90]$ (configurable knee, no longer hardcoded), and backlog backpressure multiplies the cap by $500/N_{\text{warmup}}$ beyond 500 unrefined primitives. Warmup pre-allocation is disabled (ablation −0.75 dB); microsteps fixed at $K = 5$. **Sensor-adaptive routing.** Raw depth hole-rate (missing measurements) is sensor physics, not content: TUM 0.18–0.31 vs Replica $\approx$ 0.000 ($300\text{--}500\times$). The score $n = \text{clip}(2.5 \cdot \text{holes} + 1.0 \cdot \text{edge})$ routes $n \leq 0.10 \rightarrow$ raw-error selection ($\kappa = 0.98$) and $n \geq 0.30 \rightarrow$ temporal $\times$ influence + throttling ($\kappa = 0.90$), computed pre-hygiene in milliseconds on CPU. Two earlier features failed held-out separation and are reported as such: post-hygiene invalid ratio (conflates far-but-valid geometry — 18% of Replica room0 exceeds 4 m — with holes) and Laplacian texture energy (measures sharpness, inverted: crisp synthetic scores *higher*). Operating on pre-hygiene raw depth was the fix (Fig. 1).

5 Why Ranking Fails: Negative Findings

Learned utility reverses closed-loop (−2.51 dB factorial; GradNorm beats the MLP pointwise $0.51 > 0.25$): offline labels ignore trajectory shift and inter-Gaussian coupling, so we ship no learned component. **Contextual re-ranking is unnecessary:** ground-truth rank stability $\bar{\rho} = 0.8916$ (Kendall 0.80), Top-5 overlap 80%, oracle conditional advantage exactly 0, at $424\times$ selection overhead (76.36 vs 0.18 ms). **Fixed rules are sensor-overfit**

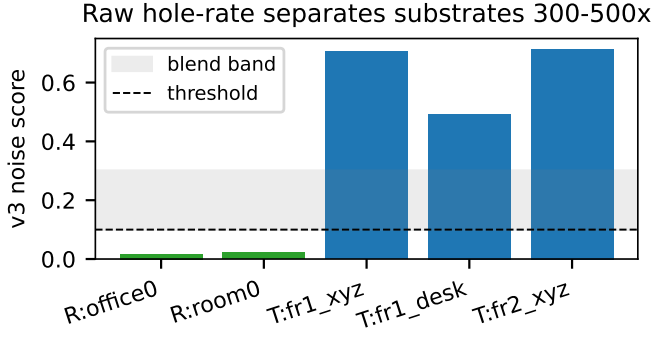


Figure 1: **Sensor separation.** v3 hole-rate score across 5 scenes: Replica ≤ 0.022 vs TUM ≥ 0.49 (thresholds 0.10/0.30, 5/5 correct). Invalid-range and texture features failed; missing-measurement rate is sensor physics, not content.

(Sec. 6): the throttled pipeline gains +3.59 dB on Kinect yet loses -3.32 dB on synthetic, where κ -ablation attributes only ~ 0.8 dB to throttling and ~ 4.7 dB to selection — the finding behind adaptive routing.

6 Experiments

Protocol. 320×240 , $B_t = 15$ ms, seeds [42..49], two-sided Wilcoxon paired + Holm-Bonferroni + bootstrap CIs; production **gsplat** backend; GT poses with tracking decoupled (mapping benchmark, not SLAM ATE); every 8th frame held out render-only (train–holdout gap $\approx +0.25$ dB, no overfitting). Artifacts: **results/phase14_corrected/**.

Main result (Tab. 1, Fig. 2, Fig. 4). OURS reaches 27.67 ± 0.04 dB: +2.63 over the strongest selective baseline, +4.62 over the RTG-SLAM-policy re-implementation, +8.80 over pass-through, all Holm-significant with $d_z \in [15.4, 168.8]$, while the unconstrained ceiling is only 0.18 dB ahead at $3.8\times$ the map.

Adaptivity is necessary (Tab. 2). On synthetic Replica the fixed throttled pipeline loses -3.32 dB; routing to raw-error selection recovers +3.32 [+3.29, +3.35], $p = 0.0078$. A κ -sweep (Fig. 3) attributes only ~ 0.8 dB of the gap to throttling; the rest is selection overfit. **Horizon effect:** at 30 frames OURS ties **error_only** (-0.01 dB on **fr2_xyz**); the lead needs ≥ 100 frames for bloat to bite — short-horizon evals hide dilution, so we benchmark 150. **Ablation (deltas transfer across substrates):** warmup -0.75 , coverage +1.04, backlog +1.08, learned utility + adaptive- K -1.54 dB. **Systems:** scheduler overhead 2.54 ms; optimization-budgeted execution at ~ 10 FPS measured — *not* end-to-end 15 ms real-time, reported as such.

7 Limitations

Ground-truth poses throughout (tracking decoupled; no ATE claim — our ICP tracker is a stub); 150-frame horizons rather than full SLAM sessions; κ /backlog thresholds calibrated on one real–synthetic pair; no learned components by

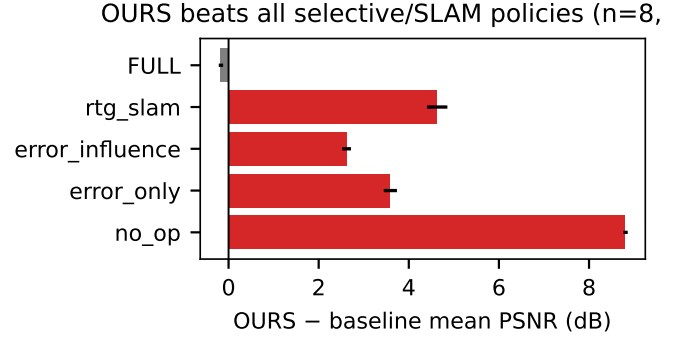


Figure 2: **Paired deltas vs OURS** with 95% bootstrap CIs (fr2_xyz, $n = 8$). All selective/SLAM baselines lose; FULL ceiling is only 0.18 dB ahead.

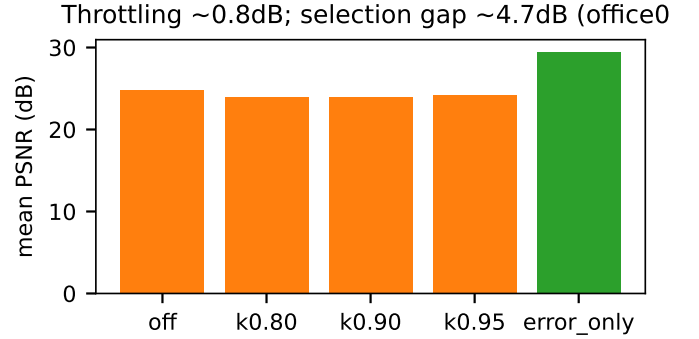


Figure 3: **Kappa ablation** (office0, $30f \times 3$): disabling throttling recovers +0.8 dB but the ~ 4.7 dB selection gap to **error_only** remains.

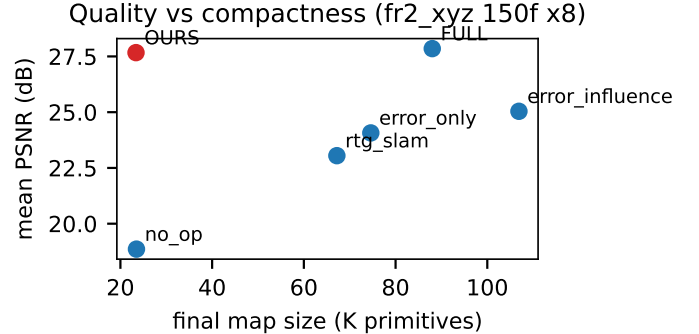


Figure 4: **Quality vs compactness.** OURS matches the ceiling at $3.8\times$ smaller map; all heuristic baselines bloat past 65k (fr2_xyz 150f $\times 8$).

design after documented failure; optimization-budgeted execution, not 15 ms end-to-end; prior 19.24 dB/76.4% numbers from a pose-bugged substrate are deprecated, not compared.

8 Conclusion

Compute dilution — not ranking error — binds budgeted online 3DGS. Dual map-growth throttling recovers 98.0% of headroom with a $3.8\times$ smaller map, and raw-hole-rate adaptive routing extends the win across noisy-real and clean-

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.86	14.88	0.6882	106.3	23,523	+8.80 ✓
error_only	24.07	20.01	0.8850	8.9	74,640	+3.59 ✓
error_influence	25.04	21.89	0.9084	8.8	106,929	+2.63 ✓
rtg_slam_reimpl	23.05	20.04	0.8663	8.0	67,231	+4.62 ✓
OURS	27.67	28.89	0.9232	10.7	23,398	—
<i>FULL</i>	<i>27.85</i>	<i>29.41</i>	<i>0.9243</i>	<i>9.5</i>	<i>87,967</i>	-0.18 ✓

Table 1: **Main benchmark, TUM fr2_xyz, 150 frames $\times$ 8 seeds.** All OURS comparisons significant ($p_{\text{Holm}} = 0.0391$). Headroom recovery $\eta = 98.0\%$ (0.18 dB gap to ceiling) at $3.8\times$ smaller map.

Same fixed policies, opposite winners: adaptivity is necessary

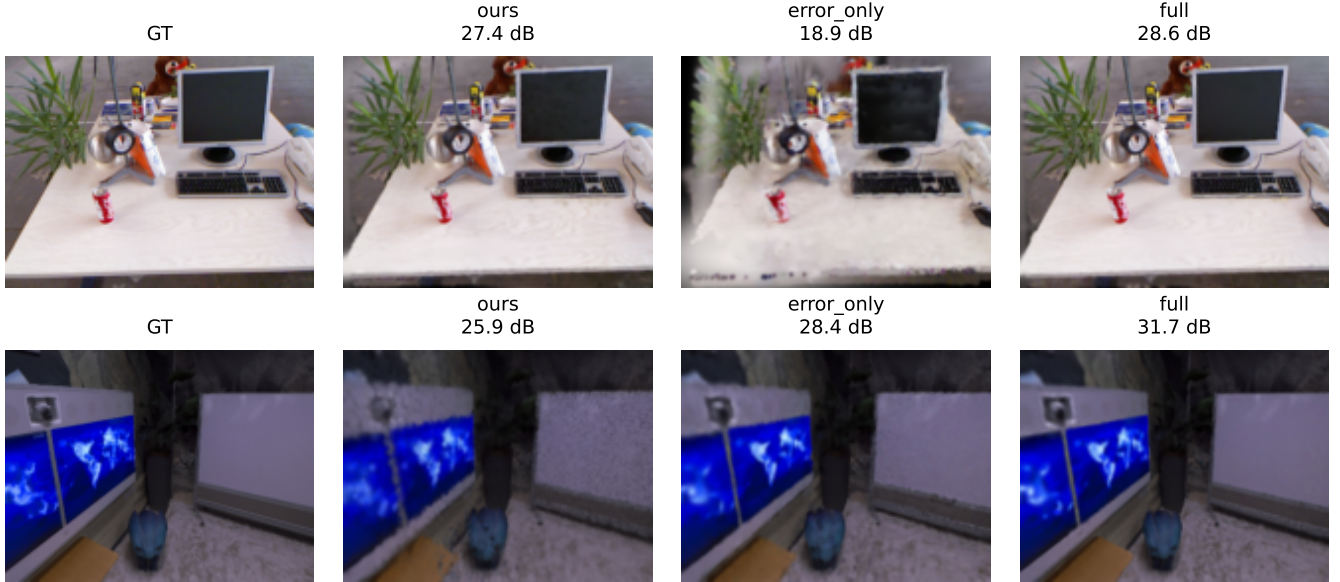


Figure 5: **Qualitative: opposite winners per substrate.** Top (TUM fr2_xyz view 149): OURS 27.4 dB matches FULL 28.6 while **error_only** collapses to 18.9. Bottom (Replica office0 view 98): **error_only** 28.4 beats OURS 25.9 near FULL 31.7. Rendered with final maps (seed 42).

Policy	Mean	Final	SSIM	Map
Fixed OURS	26.85	26.07	0.9244	78,825
Routed error_only	30.17	28.02	0.9584	115,964
Δ (adaptive – fixed)	+3.32 ✓	+1.95 ✓	+0.034	—

Table 2: **Adaptive validation, Replica office0, 150f $\times$ 8 seeds.** Router picks **error_only** (holes ≈ 0.02): $p = 0.0078$, $d_z = 71.9$. No fixed policy wins both substrates.

synthetic substrates where every fixed policy fails one. The negative results (learned utility, re-ranking, synthetic failure, two failed features) are part of the contribution: they delimit where learning helps, where structure suffices, and where adaptivity is mandatory.

References

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